Type MS · Measure food energy by burning and explain why the value is too low

Answer 1 of 8 · Question 7 part a · 6 marks

2024 NOV · Paper 01 · find part a in this mark scheme extract

7 (a)	• measure initial mass (of food)use same mass of food; • measure initial temperature of water; • burn food; • take final temperature of water; • repeat for each food; • compare temperature difference (of water)/results for each food;		6
(b)	• active transport/uptake; • reference to low to high concentration/against concentration gradient;		2

Total Marks 8

Question number	Answer	Notes	Marks

Type MS · Measure food energy by burning and explain why the value is too low

Answer 2 of 8 · Question 5 part b.i · 4 marks

2020 NOV · Paper 01 (regional R) · find part b.i in this mark scheme extract

5 (a)	(i)	**C** green vegetables and red meat		1
	(ii)	• (iron) needed for production of haemoglobin(1) • (haemoglobin/red blood cell) binds to/transports oxygen(1) • oxygen needed for (aerobic) respiration/energy(1) • for growth of fetus(1)		4
(b)	(i)	Any four from: • take temperature of water (at the start) (1) • reference to measuring mass of food(1) • place food on holder(1) • burn food(1) • reference to time (for burning food)(1) • take temperature of water after burning/once food is completely burnt/measure change in temperature(1)	Allow amount for mass Allow food is heated/set alight	4
	(ii)	Any three from: • add a lid(1) • reference to stirring/even heat distribution(1) • insulate the copper can/move food closer (to the can/water) (1) • reference to preventing heat loss(1)	Allow cover the beaker	3

Total for question 5 = 12 marks

Type MS · Measure food energy by burning and explain why the value is too low

Answer 3 of 8 · Question 3 part a.i · 5 marks

2022 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

3 (a) (i)	Any five from the following: • place water in the test tube; • take the (starting) temperature of the water; • attach the test tube to a clamp stand; • measure the mass of a piece of food/weigh food; • place food on the mounted needle (under the test tube); • burn the food; (using a Bunsen burner) • take the (final) temperature of the water;		5
(ii)	Any two from the following: • volume/mass of water; • distance of the food from the test tube; • mass of food;	ignore amount ignore amount/size	2
(iii)	Any two from the following: • hold the burning food at arm's length; • reference to sharpness of needle; • goggles; • reference to food allergies/don't taste food;	any ref to being burnt/hot water	2
(b) (i)	Any two from the following: • males need more energy than females; • the amount of energy required decreases with age; • the amount of energy needed by males decreases more (overall) than females;		2
(ii)	**D** - Males use more muscle mass than females;		1
(iii)	• (2629 – 2103) = 526; • 526 ÷ 2103; • (x 100) = 25 (%);	if divide by 2629 and give 20% then two marks Total 15 marks	3

Type MS · Measure food energy by burning and explain why the value is too low

Answer 3 of 8 · Question 3 part a.i · 5 marks

2022 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Measure food energy by burning and explain why the value is too low

Answer 4 of 8 · Question 5 part b.ii · 3 marks

2020 NOV · Paper 01 (regional R) · find part b.ii in this mark scheme extract

5 (a)	(i)	**C** green vegetables and red meat		1
	(ii)	• (iron) needed for production of haemoglobin(1) • (haemoglobin/red blood cell) binds to/transports oxygen(1) • oxygen needed for (aerobic) respiration/energy(1) • for growth of fetus(1)		4
(b)	(i)	Any four from: • take temperature of water (at the start) (1) • reference to measuring mass of food(1) • place food on holder(1) • burn food(1) • reference to time (for burning food)(1) • take temperature of water after burning/once food is completely burnt/measure change in temperature(1)	Allow amount for mass Allow food is heated/set alight	4
	(ii)	Any three from: • add a lid(1) • reference to stirring/even heat distribution(1) • insulate the copper can/move food closer (to the can/water) (1) • reference to preventing heat loss(1)	Allow cover the beaker	3

Total for question 5 = 12 marks

Type MS · Measure food energy by burning and explain why the value is too low

Answer 5 of 8 · Question 2 part a.i · 1 mark

2020 NOV · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

number				
2	(a) (i)	better transfer of heat to water (1)		1
	(ii)	any two from • safety goggles/glasses (1) • water may spit (1) • use test tube holder (1) • to avoid burning hands (1)	reason linked to precaution	2
	(b) (i)	• 28 x 12 x 4.2 (1) • 1411/1411.2J (1) • 1400J (1)	error carried forward full marks for correct answer	3
	(ii)	• some energy lost to air/absorbed by tube/needle (1) • some converted to light (1) • food not fully burnt;		3
	(c)	• use same mass of food (1) • same volume of water (1) • use same thin tube (1) • burn food same distance from tube (1)		4

Total 13 marks

Type MS · Measure food energy by burning and explain why the value is too low

Answer 5 of 8 · Question 2 part a.i · 1 mark

2020 NOV · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Measure food energy by burning and explain why the value is too low

Answer 6 of 8 · Question 2 part b.ii · 3 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

number				
2	(a) (i)	better transfer of heat to water (1)		1
	(ii)	any two from • safety goggles/glasses (1) • water may spit (1) • use test tube holder (1) • to avoid burning hands (1)	reason linked to precaution	2
	(b) (i)	• 28 x 12 x 4.2 (1) • 1411/1411.2J (1) • 1400J (1)	error carried forward full marks for correct answer	3
	(ii)	• some energy lost to air/absorbed by tube/needle (1) • some converted to light (1) • food not fully burnt;		3
	(c)	• use same mass of food (1) • same volume of water (1) • use same thin tube (1) • burn food same distance from tube (1)		4

Total 13 marks

Type MS · Measure food energy by burning and explain why the value is too low

Answer 6 of 8 · Question 2 part b.ii · 3 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Measure food energy by burning and explain why the value is too low

Answer 7 of 8 · Question 7 part a.iii · 5 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

number				
7	(a) (i)	• safety glasses/goggles; • care using sharp needle; • care to avoid burns; •		3
	(ii)	• mass of food; • volume of water; • distance of burning food from tube;	reject amount	3
	(iii)	any five from • not all food burns to release energy; • not all heat transferred to water; • some lost to atmosphere; • some heat lost to needle/glass tube; • uneven distribution of temperature in water; • because lack of stirring;		5
	(b)	any four from • use of oxygen; • allows complete combustion; • enclosed/lid so no heat escapes; • heat transfer coil so all heat is used to increase temp of water/transferred to water; • stirrer to distribute heat;		4

Total 15 marks

Type MS · Measure food energy by burning and explain why the value is too low

Answer 7 of 8 · Question 7 part a.iii · 5 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Measure food energy by burning and explain why the value is too low

Answer 8 of 8 · Question 7 part b · 4 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

number				
7	(a) (i)	• safety glasses/goggles; • care using sharp needle; • care to avoid burns; •		3
	(ii)	• mass of food; • volume of water; • distance of burning food from tube;	reject amount	3
	(iii)	any five from • not all food burns to release energy; • not all heat transferred to water; • some lost to atmosphere; • some heat lost to needle/glass tube; • uneven distribution of temperature in water; • because lack of stirring;		5
	(b)	any four from • use of oxygen; • allows complete combustion; • enclosed/lid so no heat escapes; • heat transfer coil so all heat is used to increase temp of water/transferred to water; • stirrer to distribute heat;		4

Total 15 marks

Type MS · Measure food energy by burning and explain why the value is too low

Answer 8 of 8 · Question 7 part b · 4 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks